System and Container Performance Report

1. Host-Level CPU Usage Analysis

High CPU Consuming Processes

Process	User	CPU %	Status	Notes
uvicorn	lxd	77.3%	Running	Backend API load
unattended-upgrades	root	49.1%	Disk sleep (D)	System updates running
python	lxd	100%	Running	Likely worker/job
needrestart	root	66.4%	Running	Triggered by updates
top	root	~38%	Monitoring	Expected

Observations

- CPU utilization is high, exceeding 200% combined, indicating multi-core usage.
- Background system update processes (unattended-upgrades and needrestart) are contributing to CPU load.
- Application processes (Python and Uvicorn) are consuming significant CPU resources.

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2. Container-Level Metrics Analysis

Resource Utilization Summary

Container	CPU %	Memory Usage	Memory %	PIDs	Remarks
ai-chatbot-backend-backend-1	384.95%	252.2MB / 1GB	24.62%	940	Very high CPU and process count
ai-chatbot-backend-mysql-1	209.19%	145.5MB / 512MB	28.42%	44	High database load
ai-chatbot-backend-redis-1	62.04%	3.68MB / 300MB	1.23%	6	Normal
sipcot-civil_app	0.03%	93.69MB / 31.34GB	0.29%	11	Idle
ai-chatbot-backend-weaviate-1	N/A	N/A	N/A	-	Possibly not running or unhealthy

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3. Critical Findings

3.1 Backend Container Overload

- CPU usage is extremely high at 384%.
- Process count is unusually high (940 PIDs).

Possible causes: - Excessive worker processes or threads. - Inefficient or unoptimized application logic. - High request concurrency.

This is identified as the primary bottleneck.

3.2 MySQL High CPU Usage

- CPU usage is at 209%.

Possible causes: - Heavy or inefficient queries. - Missing indexes. - Lack of query optimization.

3.3 System Update Activity

- unattended-upgrades and needrestart processes are active.
- These contribute to temporary CPU spikes.

3.4 Potential Service Issue

- The weaviate container shows no metrics.

Possible reasons: - Container is stopped or crashed. - Container is restarting or unhealthy.

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4. Overall System Health

Resource	Status
CPU	Critical (Overloaded)
Memory	Stable
Disk I/O	Moderate
Containers	Backend and database under stress

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5. Risk Summary

Risk	Impact
High CPU usage	Application slowdown or crashes

Risk	Impact
Excessive processes	System instability
Database overload	Increased API latency
System updates	Temporary performance degradation

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6. Conclusion

- The system is CPU-bound rather than memory-bound.
- The primary issue is the backend container with extremely high CPU usage and process count.
- The secondary issue is high CPU usage in the MySQL container.

Immediate optimization of backend processing and database queries is required to stabilize the system.